



CLASS PROJECT

FINAL ASSIGNMENT: BRING YOUR OWN DATA (BYOD) ▼

In the final assignment of the course you get to choose your own data as well as how to analyze it.

This assignment is composed of two steps:

- Desig your data analysis project: choosing what data to look at and describing interesting questions you'd like to answer with the data.
- Carry out the analysis.

For this final assignment you may work in teams of two or on your own.

PROJECT COMPONENTS ▼

Your project will consist of the following:

- A project proposal
- A project report in the form of the a Jupyter notebook

PROJECT PROPOSAL ▼

Your proposal should describe the topic you wish to address, the source of data you wish to analyze, what interests you about it, and the questions you would like to address with it. Provide clear steps for doing the project, and a timeline for achieving them. The proposal should be between one and two pages. You are strongly encouraged to discuss your project ideas with the instructor.

The proposal is 5% of the project grade and consists of 10 points as follows:

Grading rubric:

(4 points): You have chosen a relevant topic with an appropriate level of difficulty.

(4 pts) It is clear what the data is; you have thought of interesting questions to address using the data.

(2 points): Clear plan (what are the tasks involved and a timeline for achieving them); clear team member responsibilities for a multi-member team.

FINAL REPORT ▼

The expectations from your final report, which is submitted as a Jupyter notebook is as follows:

- The notebook should be self-contained. The work should be well-motivated and explained.
- The analysis should be non-trivial and demonstrate that you have mastered the tools we studied this semester.
- The code should make effective use of pandas and/or NumPy. It should be clear and easy to follow.
- Your are expected to visualize some aspects of the data using Matplotlib or pandas.
- Your notebook should be available via github. All data should be readable from the notebook without the need to download any files (unless the data is very large or proprietary). Submit your notebook via Canvas and provide a link to the GitHub repository at the top of the notebook.

Your report should introduce the data, its source, the questions that you are addressing with it, and contain an analysis of the data. Also include conclusions that you were able to make on the basis of your analysis. Your notebook needs to be clearly organized, self-contained, and easy to follow. It should also clearly describe what each team member did; all members of the team need to participate in creating the notebook. In grading, each team member's contribution will be taken into account.

The project will be graded based on the following criteria:

- Level of difficulty: The analysis is non-trivial, matching the level of complexity of homework assignments.
- Notebook/Writing: The notebook is self-contained with good motivation and explanation of the problem and analysis. The notebook

is well written and clear.

- Visualization: The notebook includes data visualization where appropriate; graphs are clear and convey the information effectively.
- Code: The code is correct and makes effective use of pandas and/or NumPy. It is clear and easy to follow.
- GitHub: The project is available via GitHub, including the data (as appropriate).

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